



## A new protein requirement system for dairy cows

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### ABSTRACT

Accurate prediction of protein requirements for maintenance and lactation is needed to develop more profitable diets and reduce N loss and its environmental impact. A new factorial approach for accounting for net protein requirement for maintenance ( $NP_M$ ) and metabolizable protein (MP) efficiency for lactation ( $EMP_L$ ) was developed from a meta-analysis of 223 N balance trials. We defined  $NP_M$  as the sum of the endogenous protein fecal and urinary excretion and estimated it from the intercept of a nonlinear equation between N intake and combined total N fecal and urinary excretion. Our model had a strong goodness-of-fit to estimate  $NP_M$  ( $6.32 \pm 0.15$  g protein/kg metabolic body weight;  $n = 807$  treatment means;  $r = 0.91$ ). We calculated the  $EMP_L$  as a proportion of the N intake, minus N excreted in feces and urine, that was secreted in milk. A fixed- $EMP_L$  value of  $0.705 \pm 0.020$  was proposed. In a second independent data set, nonammonia-nonmicrobial-N and microbial-N ruminal outflows were measured, and the adequacy of the MP prediction (51 studies;  $n = 192$  means treatments) was assessed. Our system based on the fixed- $EMP_L$  model predicted the MP requirement for lactation and maintenance with higher accuracy than several North American and European dairy cattle nutrition models, including the INRA (2018) and NASEM (2021). Only the NRC (2001), CNCPS 6.5, and Feed into Milk (2004) models had similar accuracy to predict MP requirement. Our system may contribute to improve the prediction for MP requirements of maintenance and lactation. However, most refined predictive models of intestinal digestibility for rumen undegradable protein and microbial protein are still needed to reduce the evaluation biases in our model and external models for predicting the MP requirements of dairy cows.

**Key words:** efficiency, maintenance, meta-analysis, metabolizable protein, modeling

### INTRODUCTION

Considerable progress has been made in the protein nutrition of dairy cows (Schwab and Broderick, 2017; Lapierre et al., 2020). However, ongoing global demand for dairy foods and high social pressure to reduce their environmental footprint continue to motivate the development of models to predict protein requirements to optimize milk production and economic performance, increase dietary N captured in milk, and minimize N excretion into the environment.

Maintenance and lactation are the two most important components of the protein requirement for lactating dairy cows. The net protein requirement for maintenance ( $NP_M$ ) has been assumed to be the sum of endogenous protein fecal and urinary excretion ( $EPFU$ ) and protein scurf losses (Owens, 1987; Lapierre et al., 2020). Invasive, expensive, and labor-intensive methods (N-free intragastric nutrition, digesta exchange, and labeling by stable isotopes) have been adopted to obtain  $EPFU$ , but they have provided limited amounts of animal and diet data (Marini et al., 2008). An alternative approach is to estimate  $EPFU$  by extrapolation of the intercept-regression between N intake and N total fecal and urinary excretion ( $NFU$ ; Swanson, 1977; Owens, 1987). As balance N trials have been widely published, meta-analyses of N balance trials may provide a robust estimate of  $EPFU$  and  $NP_M$ .

Several dairy cow nutrition models (NRC, 2001; Van Duinkerken et al., 2011; Van Amburgh et al., 2015) have adopted  $EPFU$  from Swanson's (1977) meta-analysis of N metabolism trials of nonlactating cattle. As lactating dairy cows typically have higher N intake and excretion per body mass than nonlactating cattle, the Swanson (1977) model may underestimate  $NP_M$ . The NASEM (2021) committee adopted the assumptions of Lapierre et al. (2020) to predict  $EPFU$  from an admittedly scarce literature review of dairy cows. The new NASEM (2021) model seems to have improved the prediction of milk protein yield compared with NRC (2001), but an evaluation of its modeled protein requirement against observed values was not reported.

The net protein requirement for lactation ( $NP_L$ ) is more easily determined and represents the milk true protein secretion. However, to obtain the MP require-

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ment for lactation ( $MP_L$ ), it is necessary to know the efficiency of the use of MP to  $NP_L$  ( $EMP_L$ ). Most dairy cattle nutrition committees have adopted fixed- $EMP_L$  values to predict  $MP_L$  from  $NP_L$ , with  $EMP_L$  ranging from 0.67 to 0.70 (NRC, 2001; CSIRO, 2007; Van Amburgh et al., 2015; INRA, 2018; NASEM 2021). The INRA (2018) and NASEM (2021) models adopted variable- $EMP_L$ -based models to predict MP supply and milk protein yield, but fixed- $EMP_L$  values of 0.67 and 0.69 were used as targets to predict  $MP_L$  requirements from  $NP_L$ .

To the best of our knowledge, only the Dutch protein evaluation system for ruminants (**DVE/OEB**<sub>2010</sub>; Van Duinkerken et al., 2011) and NorFor (2011) models have used variable- $EMP_L$  to predict  $MP_L$  from  $NP_L$  requirements, using the milk protein to energy ratio and the MP supply available for milk production to milk energy ratio as inputs, respectively. In DVE/OEB<sub>2010</sub>,  $EMP_L$  decreases with milk protein yield (Van Duinkerken et al., 2011), whereas in NorFor (2011) neither the milk protein yield nor the feeding level affects  $EMP_L$ . However, sufficient evidence exists to show that milk yield and N milk efficiency (N milk/N intake) are positively correlated (Nadeau et al., 2007), but a better prediction of the MP requirement using a variable- $EMP_L$ -based model instead of fixed- $EMP_L$  remains unclear.

A more comprehensive and accurate system to predict NP and MP requirements for lactating cows is necessary. We hypothesized that (1) a new model from a meta-analysis of N balance trials may provide a robust estimate of EPFU,  $NP_M$ , and  $EMP_L$ ; and (2) our new system may improve MP requirement prediction for lactating dairy cows compared with external dairy cattle nutrition models.

Our objectives were (1) to propose new values for EPFU ( $NP_M$ ) and  $EMP_L$  for lactating dairy cows from a N balance trials meta-analysis, and (2) to compare the adequacy of our protein requirements system with external models to predict the MP requirement for lactating dairy cows. The proposed protein requirement will be used to update the protein submodel of the Nutrition System for Dairy Cattle (Oliveira, 2019).

## MATERIALS AND METHODS

Institutional Animal Care and Use Committee approval was not necessary, because our data set was built from a systematic review of peer-reviewed papers.

### Protein Requirement System Development

**Data Set.** A large data set of N balance trials was built to develop our protein requirement system, based on a search of peer-reviewed papers using the terms

“nitrogen” and “dairy cows” in the Web of Science and Science Direct databases. The literature search yielded 5,418 peer-reviewed papers. The inclusion criteria for the data set were (1) peer-reviewed papers; (2) studies conducted with lactating cows; (3) reporting of the treatment means for CP intake, milk protein yield, and N excretion in feces and urine; and (4) reporting of the standard error of the mean (**SEM**) or the standard error of the difference (**SED**). When SED was reported in studies evaluated as fixed models, SEM was calculated as  $SEM = SED/\sqrt{2}$ . A flowchart showing the process of identification, exclusion, and inclusion of studies to construct the protein requirement model is described in Supplemental Material S1 (<https://data.mendeley.com/datasets/z6pdk5pyg9>; Silva and Oliveira, 2022b).

Based on the inclusion criteria, 212 peer-reviewed papers involving 223 N balance trials were selected for data extraction and derivation of our protein requirement system (Table 1). No procedure was adopted to estimate missing data, except for the SEM of N urinary excretion. Data that did not report on studies were considered as missing and subsequently excluded from the final model. No adjustment was made to fecal N obtained from oven-dried samples (Juko et al., 1961; Higgs et al., 2012). The complete data set in the Excel file is available in an open research data repository (Silva and Oliveira, 2022a), and the references used to develop the models are available in Appendix 1 of Supplemental Material S2 (<https://data.mendeley.com/datasets/z6pdk5pyg9>; Silva and Oliveira, 2022b).

Each observation (treatment means) was classified by genetic group (*Bos taurus* or *B. taurus* × *Bos indicus*), feed system (TMR or pasture), DIM group (<100 DIM or ≥100), fecal output method (total collection or fecal marker), and urinary output method (total collection or spot sample using creatinine as marker) for use as potential covariates in models.

**Data Weighting.** Each observation (treatment means) used to derive our models was weighted by the normalized inverse of the SEM (St-Pierre, 2001) of N urinary excretion (g/d), as follows. The normalized inverse of the SEM of N urinary excretion was calculated as  $W_1/W_2$ , where  $W_1 = 1/\text{reported SEM of N urinary excretion (g/d)}$  of each observation/study (g/d), and  $W_2 = \text{overall mean of } W_1 \text{ across studies}$ . To prevent overweighting of studies with extremely low SEM (Elliott, 2008), we truncated (i.e., trimmed) the SEM to  $0.35 \times \text{overall mean SEM}$  (Roman-Garcia et al., 2016) of N urinary excretion. This procedure was conducted separately for the studies that adopted mixed and fixed effects models because mixed models tend to have higher SEM (Littell et al., 1998). Missing data of SEM of N urinary (n = 104 observations not reported in

**Table 1.** Descriptive statistics of the data set used to develop the new protein requirement system for dairy cows

| Item                               | Mean  | Median | Maximum | Minimum | SD    | n <sup>1</sup> |
|------------------------------------|-------|--------|---------|---------|-------|----------------|
| Animal                             |       |        |         |         |       |                |
| BW, kg                             | 616   | 622    | 788     | 351     | 66.9  | 856            |
| Milk yield, kg/d                   | 30.8  | 30.3   | 50.1    | 9.02    | 8.40  | 784            |
| DIM                                | 113   | 103    | 323     | 14.0    | 55.5  | 653            |
| Milk protein, g/kg                 | 31.7  | 31.3   | 39.6    | 25.7    | 2.29  | 573            |
| Milk fat, g/kg                     | 37.9  | 37.5   | 60.8    | 22.7    | 5.19  | 743            |
| Milk lactose, g/kg                 | 47.5  | 47.7   | 54.0    | 41.0    | 1.74  | 571            |
| Milk urea-N, mg/dL                 | 13.1  | 12.7   | 34.0    | 3.66    | 4.17  | 425            |
| Diet                               |       |        |         |         |       |                |
| DMI, kg/d                          | 21.0  | 21.2   | 31.8    | 9.20    | 4.16  | 784            |
| Forage, g/kg DM                    | 539   | 533    | 1,000   | 218     | 91.5  | 507            |
| DM, g/kg                           | 548   | 537    | 900     | 317     | 101   | 292            |
| OM, g/kg DM                        | 928   | 927    | 994     | 874     | 18.5  | 265            |
| NDF, g/kg DM                       | 339   | 334    | 534     | 205     | 60.7  | 525            |
| CP, g/kg DM                        | 167   | 165    | 281     | 94.0    | 21.5  | 562            |
| Ether extract, g/kg DM             | 38.7  | 37.2   | 89.0    | 15.0    | 12.4  | 302            |
| Starch, g/kg DM                    | 217   | 226    | 359     | 33.9    | 67.2  | 241            |
| NFC, g/kg DM                       | 416   | 418    | 550     | 101     | 63.2  | 249            |
| DM total-tract digestibility       | 0.675 | 0.677  | 0.834   | 0.482   | 0.047 | 545            |
| OM total-tract digestibility       | 0.697 | 0.700  | 0.848   | 0.571   | 0.044 | 466            |
| NE <sub>L</sub> diet, Mcal/kg DM   | 1.60  | 1.60   | 1.78    | 1.37    | 0.07  | 275            |
| N balance and efficiency           |       |        |         |         |       |                |
| N intake, g/d                      | 550   | 557    | 954     | 183     | 134   | 856            |
| N milk, g/d                        | 147   | 148    | 247     | 46.6    | 39.6  | 856            |
| N urinary, g/d                     | 184   | 182    | 437     | 15.3    | 68.9  | 856            |
| N-urea urinary, g/d                | 160   | 143    | 663     | 41.0    | 89.2  | 159            |
| N fecal, g/d                       | 184   | 176    | 386     | 34.4    | 55.8  | 856            |
| SEM of N urinary, <sup>2</sup> g/d | 10.2  | 9.20   | 54.8    | 1.49    | 5.67  | 752            |
| N milk efficiency <sup>3</sup>     | 0.27  | 0.27   | 0.47    | 0.11    | 0.05  | 856            |

<sup>1</sup>Treatment means from 223 N balance trials in 212 peer-reviewed papers. The complete data set is available in an Excel file in Silva and Oliveira (2022a). The list of references is available in Supplemental Material S2 (<https://data.mendeley.com/datasets/z6pdk5pyg9>; Silva and Oliveira, 2022b). Missing data represent non-reported data in studies. No missing data of response or predictor variables were estimated, except for the SEM of N urinary.

<sup>2</sup>The SEM of N urinary excretions reported in studies. The SEM of N urinary was not reported in 104 observations (missing data). On final models, these missing data were estimated using observed overall means across studies and evaluated separately by mixed and fixed models, after the SEM truncate procedure.

<sup>3</sup>N milk (g/d)/N intake (g/d).

studies; Table 1) were estimated using observed overall mean across studies, evaluated separately by mixed and fixed models, and after the SEM truncation procedure.

**Net Protein Requirement for Maintenance.** We assumed that NP<sub>M</sub> is the EPFU. Scurf protein losses (skin and hair scaling) were not accounted for in NP<sub>M</sub> in our system for 2 reasons. First, data were absent from our data set. Second, scurf protein represents only 1.8% (range 1.4–2.6%) of the sum of the EPFU, calculated from Lapierre et al. (2020) and assuming dairy cows with 616 kg of BW, 21 kg/d DMI, and 34% NDF in DM diet (Table 1).

The net protein requirement for maintenance was modeled as 6.25 (factor N-protein) × intercept of the regression between N intake [g of N/kg metabolic body weight ( $\text{BW}^{0.75}$ ); predictor variable] and NFU (g of N/kg of  $\text{BW}^{0.75}$ ; response variable), using a nonlinear mixed model and adaptive Gaussian quadrature as the integration method, as follows:

$$Y_{ij} = \beta_1 \times \exp(\text{N intake} \times \beta_2) + \text{trial}_j + e_{ij}, \quad [1]$$

where  $Y_{ij}$  = NFU of the treatment means  $i$  of the N balance trial  $j$ ;  $\beta_1$  = overall intercept across all studies (fixed effects), and it represents the sum of the endogenous fecal and urinary N excretion (g of N/kg of  $\text{BW}^{0.75}$ );  $\beta_2$  = overall nonlinear statistics across all trials (fixed effect), without nutritional significance;  $\text{trial}_j$  = random effect of N balance trial  $j$ ; and  $e_{ij}$  = random error associated with each observation, assuming a normal distribution (0,  $\sigma^2$ ). The Kleiber's 0.75 inter-specific body mass exponent was adopted for N intake (g of N/kg of  $\text{BW}^{0.75}$ ) and NFU (g of N/kg of  $\text{BW}^{0.75}$ ). Although several conversion factors of N-protein have been proposed for feedstuffs (Mariotti et al., 2008), we adopted the classical Jones conversion factor of 6.25 for the EPFU (NASEM, 2016, 2021; Lapierre et al., 2020). To calculate the MP requirement for maintenance from

$NP_M$ , we assumed that maintenance has the same MP efficiency for lactation.

We initially evaluated the interaction effect of feed system, genetic group, DIM group, fecal output method, and urinary output method with the intercept, using a linear multivariable mixed model with a variance component structure (St-Pierre, 2001). If these interaction effects had a  $P$ -value  $> 0.10$ , an overall nonlinear mixed Equation 1 without covariates was proposed to obtain  $NP_M$ .

Observations were weighted by normalized inverse of the SEM of N urinary excretion (g/d), as previously described. Observations were removed if the studentized residual was outside the range of  $-2.0$  to  $2.0$ . A list of removed observations ( $n = 49$ ) is available in Silva and Oliveira (2022a). Significance was declared at  $P \leq 0.05$ . Analyses were conducted using the PROC MIXED and PROC NLMIXED procedures (Littell et al., 2006) of SAS OnDemand for Academics (SAS Institute Inc.). As the WEIGHT statement is not available in the PROC NLMIXED procedure, the REPLICATE statement was adopted as a WEIGHT statement (SAS Institute Inc., 2015). The SAS codes and outputs are available in Supplemental Material S5 (<https://data.mendeley.com/datasets/z6pdk5pyg9>; Silva and Oliveira, 2022b).

**Efficiency of MP Utilization for Lactation.** We calculated the  $EMP_L$  (0–1) as a proportion of the N intake that was secreted in milk, discounting N excreted in feces and urine, as follows:

$$EMP_L = \frac{N_{\text{milk}} \left( \text{g/kg BW}^{0.75} \right) \times TP/CP}{N_{\text{intake}} \left( \text{g/kg BW}^{0.75} \right) - N_{\text{fecal}} \left( \text{g/kg BW}^{0.75} \right) - N_{\text{urinary}} \left( \text{g/kg BW}^{0.75} \right)}, \quad [2]$$

where  $TP/CP$  is the ratio of true protein to CP in milk and is equal to 0.955 (Moraes et al., 2018). We also initially analyzed the interaction effect of feed system, genetic group, DIM group, fecal output method, and urinary output method on  $EMP_L$ , using a multivariable mixed model with a component variance structure (St-Pierre, 2001), considering N balance trial as a random effect and covariates as fixed effects, similar to that adopted for the NFU model.

We proposed a fixed- $EMP_L$  model for predicting MP requirements. The fixed- $EMP_L$  value represented the least squares means obtained from the estimated  $EMP_L$  (Equation 2) of each observation (treatment mean), using a mixed model with N balance trial as a random effect. Observations used to obtain the fixed  $EMP_L$  were weighed by the normalized inverse of the SEM of N urinary excretion (g/d). Observations were removed

if the studentized residual was outside the range of  $-2.0$  to  $2.0$ . A list of removed observations ( $n = 112$ ) is available in Silva and Oliveira (2022a). Significance was declared at  $P \leq 0.05$ . Analyses were conducted using the PROC MIXED procedures (Littell et al., 2006) of SAS OnDemand for Academics.

### MP Model Evaluations

To evaluate the prediction of the MP requirement (maintenance + lactation) of our proposed model and eight external models (Supplemental Material S3; <https://data.mendeley.com/datasets/z6pdk5pyg9>; Silva and Oliveira, 2022b), we built an independent data set of 51 trials (from 51 peer-reviewed papers) that measured total nonammonia-N and microbial-N ruminal outflows in lactating dairy cows from duodenal or omasal digesta sampling ( $n = 192$  treatment means; Table 2). Nonammonia nonmicrobial-N ruminal outflow (NANMN; as a proxy for RUP) was obtained from difference between total NAN and microbial-N ruminal outflows. Total nonammonia-N and microbial-N ruminal outflows were measured from duodenal (41 studies) or omasal (10 studies) digesta sampling, using purine (31 studies),  $^{15}\text{N}$  label (14 studies), diaminopimelic acid (5 studies), or RNA-cytosine (1 study) for microbial markers. The complete data set in the Excel file is available in an open research data repository (Silva and Oliveira, 2022a), and references used to evaluate the models are available in Appendix 2 of Supplemental Material S2 (<https://data.mendeley.com/datasets/z6pdk5pyg9>; Silva and Oliveira, 2022b).

The observed MP of each treatment mean was calculated as follows:

$$\begin{aligned} \text{Observed MP (g/cow/d)} = & \text{NANMN (g/cow/d)} \times 6.25 \times \text{ID-RUP} \\ & + \text{microbial-N ruminal outflow (g/cow/d)} \times 6.25 \\ & \times \text{TP/CP microbial} \times \text{ID-TP microbial} \end{aligned} \quad [3]$$

where 6.25 is the conversion factor N to protein; ID-RUP = overall intestinal digestibility of the RUP of 0.79, calculated from intestinal digestibility of total AA of 25 feeds reported in Supplementary Table S4 of White et al. (2017);  $TP/CP$  microbial = true protein to CP ratio in microbial protein of 0.82 (Sok et al., 2017); and ID-TP microbial = overall intestinal digestibility of the true protein microbial of 0.80 (NASEM, 2021). Endogenous protein duodenal flux was not calculated as observed MP, according to Lapierre et al. (2020).

We compared the adequacy of the MP requirement prediction of our proposed model with several exter-



**Table 2.** Descriptive statistics of the complete data set used to evaluate MP requirements models for lactating dairy cows

| Item                             | Mean  | Median | Maximum | Minimum | SD   | n <sup>1</sup> |
|----------------------------------|-------|--------|---------|---------|------|----------------|
| Animal                           |       |        |         |         |      |                |
| BW, kg                           | 601   | 598    | 736     | 480     | 59.2 | 192            |
| Milk yield, kg/d                 | 28.6  | 28.3   | 48.2    | 13.0    | 7.1  | 192            |
| DIM                              | 114   | 110    | 247     | 32      | 55   | 170            |
| Milk protein, g/kg               | 31.4  | 31.3   | 37.4    | 25.9    | 2.65 | 192            |
| Milk fat, g/kg                   | 35.0  | 35.1   | 46.7    | 22.9    | 4.34 | 192            |
| Milk urea N, mg/dL               | 13.6  | 13.3   | 23.5    | 4.87    | 4.60 | 29             |
| Diet                             |       |        |         |         |      |                |
| DMI, kg/d                        | 20.4  | 20.7   | 31.6    | 9.11    | 3.88 | 174            |
| Forage in diet, g/kg DM          | 489   | 487    | 800     | 250     | 101  | 123            |
| DM, g/kg                         | 576   | 575    | 782     | 304     | 95.8 | 93             |
| OM, g/kg DM                      | 919   | 919    | 947     | 811     | 15.3 | 83             |
| NDF, g/kg DM                     | 343   | 342    | 623     | 220     | 58.7 | 182            |
| CP, g/kg DM                      | 170   | 171    | 212     | 113     | 17.0 | 139            |
| Ether extract, g/kg DM           | 42.2  | 42.5   | 55.6    | 28.1    | 9.94 | 24             |
| Starch, g/kg DM                  | 269   | 282    | 476     | 32.9    | 105  | 63             |
| DM digestibility, g/kg           | 664   | 670    | 730     | 582     | 34.1 | 89             |
| OM digestibility, g/kg           | 675   | 673    | 772     | 557     | 41.3 | 160            |
| NE <sub>L</sub> diet, Mcal/kg DM | 1.68  | 1.67   | 1.86    | 1.54    | 0.07 | 68             |
| N intake and efficiency          |       |        |         |         |      |                |
| N intake, g/d                    | 553   | 567    | 769     | 246     | 112  | 191            |
| N milk, g/d                      | 141   | 138    | 229     | 64.9    | 34.3 | 192            |
| N milk efficiency <sup>2</sup>   | 0.26  | 0.26   | 0.36    | 0.16    | 0.04 | 189            |
| N milk/DMI, g/kg DM              | 7.08  | 7.01   | 10.2    | 4.25    | 1.03 | 174            |
| Ruminal N outflow <sup>3</sup>   |       |        |         |         |      |                |
| NAN, g/d                         | 513.5 | 503.9  | 858     | 246     | 132  | 192            |
| Microbial N, g/d                 | 288.4 | 268.8  | 570     | 136     | 90.0 | 192            |
| NANMN, g/d                       | 225.1 | 222.5  | 576     | 75.0    | 88.0 | 192            |
| MP, <sup>4</sup> g/d             | 2,294 | 2,236  | 3,880   | 1,096   | 597  | 192            |

<sup>1</sup>Treatment means from 51 studies in 51 peer-reviewed papers. The complete data set is available in an Excel file in Silva and Oliveira (2022a). List of reference is available in Supplemental Material S3. Missing data (nonreported in studies) of DMI (n = 18) were estimated from the NRC (2001) equation because DMI is an input for the proposed Model II and several external models (Supplemental Material S3, <https://data.mendeley.com/datasets/z6pdk5pyg9>; Silva and Oliveira, 2022b). Missing data (nonreported in studies) of NDF (n = 10), total-tract DM digestibility (n = 103), and OM digestibility (n = 32) were estimated using observed overall mean values across studies because they also are inputs for some external models (Supplemental Material S3).

<sup>2</sup>N milk efficiency = N milk (g/d)/N intake (g/d).

<sup>3</sup>NAN = nonammonia N ruminal outflow; microbial n = microbial nitrogen ruminal outflow; NANMN = non-ammonia nonmicrobial N ruminal outflow.

<sup>4</sup>Observed MP was calculated according to Equation 3.

nal models: AFRC (1993), NRC (2001), Feed into Milk (Thomas, 2004), CSIRO (2007), DVE/OEB<sub>2010</sub> (Van Duinkerken et al., 2010), CNCPS 6.5 (Fox et al., 2004; Van Amburgh et al., 2015), INRA (2018), and NASEM (2021). The description of the external models is available in Supplemental Material S3 (<https://data.mendeley.com/datasets/z6pdk5pyg9>; Silva and Oliveira, 2022b). The NorFor (2011) model was not evaluated because it requires inputs that were not reported in studies used in our data set (Silva and Oliveira, 2022a).

Missing data (nonreported in studies) of DMI (n = 18) were estimated from the NRC (2001) equation because DMI is an input for several external models (Supplemental Material S3, <https://data.mendeley.com/datasets/z6pdk5pyg9>; Silva and Oliveira, 2022b). Missing data (nonreported in studies) of NDF (n = 10), total-tract DM digestibility (n = 103), and OM digest-

ibility (n = 32) were estimated using observed overall mean value across studies because they also are inputs for some external models (Supplemental Material S3).

The values of observed MP (Equation 4) of each treatment mean (Table 2) were each compared with its respective predicted MP requirements (maintenance + lactation) values from each model (proposed and external). The adequacy of the MP requirement predictive models was assessed for precision and accuracy by simple linear regression of the observed MP values (Y) with the predicted MP requirement (X), using the following procedures: graphical analysis of observed versus predicted values and its residuals, coefficient of determination (R<sup>2</sup>), mean square of prediction error (MSPE) and its decomposition in 3 sources of error (error caused by the bias, error caused by the deviation of regression slope from unity, and random error; Theil,

1966; Bibby and Toutenburg, 1977), and concordance correlation coefficient (CCC) and its decomposition into precision ( $\rho$ , correlation coefficient) and accuracy ( $C_b$ , bias correction factor) indicators (Lin, 1989).

The slope and intercept between residuals and predicted MP requirements of all models were tested to quantify the magnitude of the mean bias and linear bias centralized to their mean values, respectively (St-Pierre, 2003). An ANOVA component was conducted to identify potential effects of animal performance (DMI, milk protein yield, and BW) and dietary (NDF and CP content, and OM total-tract digestibility) variables, and digesta sampling method (duodenal versus omasal) affecting MP residual (observed minus predicted MP) of our proposed model, using a linear multivariable mixed model with variance component structure (Littell et al., 2006).

In addition, to compare accuracy between the models, the root MSPE (RMSPE) was calculated as  $RMSPE = \sqrt{[\sum SPE/(n - 2)]}$ , and its standard error  $[SE_{(SPE)}]$  was calculated as  $SE_{(SPE)} = \sqrt{SD_{(SPE)}/\sqrt{(n)}}$ , where SPE = square prediction error = (observed MP – predicted MP requirement)<sup>2</sup>, n = number of observations (treatment means used to independently evaluate the models), and  $SD_{(SPE)}$  = standard deviation of the SPE.

## RESULTS

### Proposed Protein Requirement System

**Data Set.** The complete data set used to develop our protein requirement system for lactating dairy cows was from 26 countries and represented a wide range of animal performance (milk yield of 9–50 kg/d; BW of 351–788 kg), diet (218–1,000 g forage/kg DM; 94–281 g CP/kg DM), and N metabolism (Table 1). The United States was the principal country of origin (38.1%), followed by Canada (16.5%), the United Kingdom (6.9%), Brazil (5%), Finland (5%), France (3.5%), Switzerland (3.5%), and 20 other countries (21.5% data set). Rotational trials were the most common experimental design used (81.7%), and *B. taurus* (Holstein) was the main genotype (96.3%). The predominant feeding system was TMR (91.2%), and DIM  $\geq 100$  d represented 56.4% of the data set. Forages were used as silage (54.8%), haylage (27.8%), hay (6.1%), pasture (8.8%), and other (2.5%). Total collection for quantifying fecal and urinary N excretion was used in 55.8% and 59.3% of studies, respectively.

**Protein Requirement for Maintenance.** Feed system (TMR or pasture;  $P = 0.28$ ), genetic group (*B. taurus* or *B. taurus*  $\times$  *B. indicus*;  $P = 0.15$ ), lactation stage ( $<100$  DIM or  $\geq 100$  DIM;  $P = 0.38$ ), fecal output method (total collection or fecal marker;  $P = 0.94$ ),

and urinary output method (total collection or spot;  $P = 0.61$ ) did not affect endogenous NFU (intercept between N intake and sum of the total fecal and urinary excretion; Table 3). Therefore, we proposed to use one overall equation to predict endogenous NFU and  $NP_M$  (Figures 1 and 2):

Endogenous NFU =

$$1.012 \pm 0.024 \text{ g of N} \times \text{kg of BW}^{0.75}, \quad [4]$$

$$NP_M = 6.32 \pm 0.15 \text{ g of protein} \times \text{kg of BW}^{0.75}. \quad [5]$$

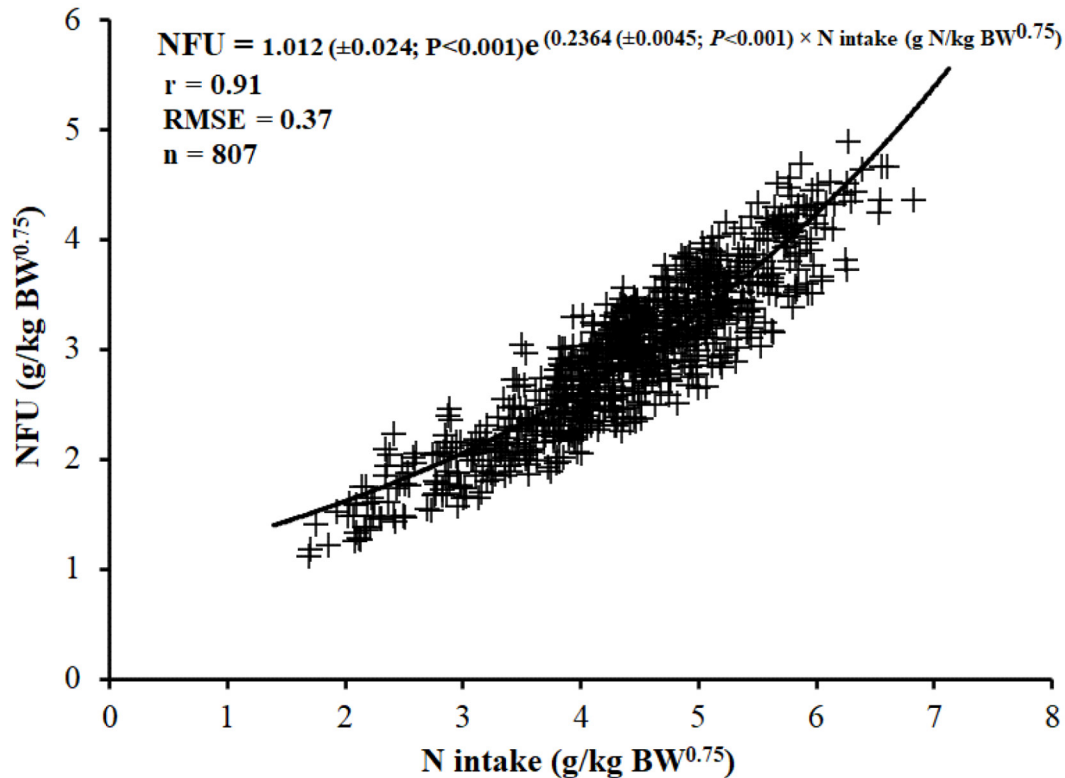
Nitrogen fecal and urinary excretion (g of N/BW<sup>0.75</sup>) had a strong correlation ( $r = 0.91$ ) with N intake (g of N/BW<sup>0.75</sup>) (Figures 1 and 2). Study ( $P = 0.24$ ) and N intake  $\times$  study ( $P = 0.79$ ) did not affect NFU, and variance of study (between-study heterogeneity) contributed less than 1% of the total variance of NFU (Figure 2).

**Efficiency of MP for Lactation.** The feed system, DIM, fecal output method, and urinary output method did not affect  $EMP_L$  (Table 3), but the genetic group affected ( $P = 0.01$ )  $EMP_L$  (Table 3). Therefore, fixed- $EMP_L$  values were obtained for each genetic group: *B. taurus*  $EMP_L = 0.705 \pm 0.020$  [ $n = 705$ ; milk yield = 10–50 kg/d, BW = 351–788 kg, DMI = 9.2–31.8 kg/d, and N milk efficiency = 0.11–0.27 (mean  $0.27 \pm 0.05$ ); Table S4.1 of Supplemental Material S4, <https://data.mendeley.com/datasets/z6pdk5pyg9>; Silva and Oliveira, 2022b]; and *B. taurus*  $\times$  *B. indicus*  $EMP_L =$

**Table 3.** ANOVA of the effect of feeding system, genetic group, lactation stage, fecal output method, and urinary output method on the sum of N total fecal and urinary excretion (NFU), and efficiency of MP utilization for lactation ( $EMP_L$ ) of dairy cows

| Effect                                                                               | NFU   | $EMP_L$ |
|--------------------------------------------------------------------------------------|-------|---------|
| Solution for fixed effects <sup>1</sup>                                              |       |         |
| Intercept (estimated endogenous NFU), g/kg BW <sup>0.75</sup>                        | <0.01 |         |
| Nitrogen intake (NI), g/kg BW <sup>0.75</sup>                                        | <0.01 |         |
| Feed system (TMR or pasture)                                                         | 0.28  | 0.17    |
| Genetic group ( <i>Bos taurus</i> or <i>Bos taurus</i> $\times$ <i>Bos indicus</i> ) | 0.15  | 0.01    |
| Lactation stage ( $\leq 100$ DIM or $>100$ DIM)                                      | 0.38  | 0.15    |
| Fecal output method (total collection or fecal marker)                               | 0.94  | 0.83    |
| Urinary output method (total collection or spot)                                     | 0.61  | 0.46    |
| Feed system $\times$ NI                                                              | 0.40  |         |
| Genetic group $\times$ NI                                                            | 0.10  |         |
| Lactation stage $\times$ NI                                                          | 0.81  |         |
| Fecal output method $\times$ NI                                                      | 0.93  |         |
| Urinary output method $\times$ NI                                                    | 0.80  |         |
| Observations used                                                                    | 571   | 548     |

<sup>1</sup>Using mixed model with random effect of study and variance component structure, weighed by normalized inverse of SEM of the N urinary excretion (g/d). The results were from multivariable model.



**Figure 1.** Relationship between the sum of N total fecal and urinary excretion (NFU; g of N/kg of  $BW^{0.75}$ ) and N intake (g of N/kg of  $BW^{0.75}$ ) from 807 means treatment (experimental diets) of 199 N balance trials. Endogenous NFU excretion represents the intercept of the regression between NFU and N intake ( $1.012 \pm 0.024$  g of N  $\times$  kg of  $BW^{0.75}$ ). Net protein for maintenance ( $NP_M$ ; g of protein/kg of  $BW^{0.75}$ ) =  $6.25 \times$  endogenous NFU excretion (g of N/kg of  $BW^{0.75}$ ); therefore  $NP_M = 6.32 \pm 0.15$  g of protein  $\times$  kg of  $BW^{0.75}$ . RMSE = root mean square error. In the total 856 observations available (Table 1), 49 were removed from analysis of studentized residues (outliers). The complete data set and a list of the 49 observations evaluated as outliers are available in Silva and Oliveira (2022a). The SAS (SAS OnDemand for Academics, SAS Institute Inc.) codes and outputs to derive the model are available in Supplemental Material S5.1 (<https://data.mendeley.com/datasets/z6pdk5pyg9>; Silva and Oliveira, 2022b).

$0.594 \pm 0.045$  [ $n = 26$ ; milk yield = 9–21 kg/d,  $BW = 389$ –580 kg,  $DMI = 11.5$ –20.3 kg/d, and N milk efficiency =  $0.16$ – $0.30$  (mean  $0.24 \pm 0.03$ ); Table S4.2 of Supplemental Material S4). However, the  $EMP_L$  for *B. taurus*  $\times$  *B. indicus* should be interpreted with caution owing to the low number of treatments means ( $n = 26$ ) compared with *B. taurus* ( $n = 705$ ; mainly Holstein).

**Protein Requirement System.** We proposed a model to predict MP requirements (maintenance and lactation) for dairy cows using a fixed value of MP efficiency for maintenance and lactation (Table 4). We assumed that maintenance (combined EPFU) has the same MP efficiency as lactation (Table 4).

### Model Evaluation of Protein Requirement

**Proposed Model.** Our proposed model predicted the MP requirement (maintenance + lactation) with a nonsignificant mean bias (observed minus predicted;  $-4.0 \pm 32.8$  g of MP/d;  $P = 0.90$ ; and  $0.0\%$  MSPE) and linear (slope) bias (slope =  $0.19 \pm 0.10$ ;  $P = 0.07$ ;  $5.8\%$

MSPE), and with high random error (not explained by the model;  $94.1\%$  MSPE; Table 5 and Figure 3). The model predicted with  $\rho = 0.65$  (precision),  $C_b = 0.84$  (accuracy),  $CCC = 0.55$  (combined precision and accuracy), and  $RMSPE = 20.0 \pm 1.7\%$  (observed MP  $\pm$  SE; Figure 5).

Among the evaluated variables,  $DMI$  ( $15.6\%$  total variance),  $OM$  total-tract digestibility ( $10.7\%$ ), and  $CP$  diet ( $10.1\%$ ) had the greatest effect ( $P < 0.01$ ) on the variance of the MP residual of the proposed model (Table 6). The increase of  $DMI$  and  $CP$  diet increased the MP residual, while the increase of the  $OM$  total-tract digestibility reduced the MP residual (Table 6). Milk protein yield and  $NDF$  diet also affected the variance of the MP residual, but at a lower magnitude ( $3.1$  and  $3.3\%$  total variance of the MP residual; Table 6). Digesta sampling method (duodenal vs. omasal) did not affect the MP residual of the proposed model (Table 6).

**Comparison of the Proposed Model with External Models.** The AFRC (1993), CSIRO (2007), DVE/OEB<sub>2010</sub> (Van Duinkerken et al., 2011), INRA

**Table 4.** Description of the proposed system of net protein (NP) and MP requirements for maintenance and lactation of dairy cows

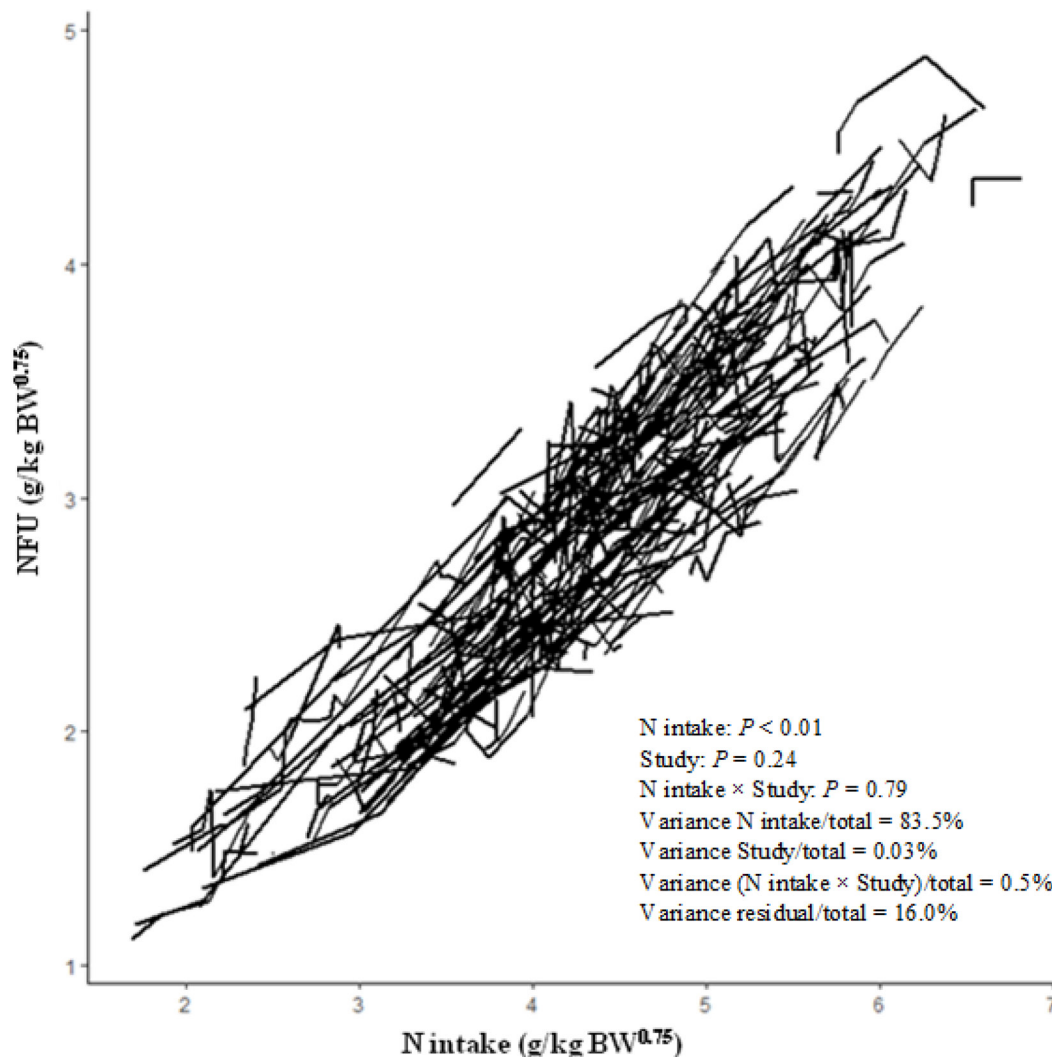
| Item                                                                  | Proposed model ( $\pm$ SE) <sup>1</sup>   |
|-----------------------------------------------------------------------|-------------------------------------------|
| Net protein requirement for maintenance (NP <sub>M</sub> ), g/d       | $(6.32 \pm 0.15) \times \text{BW}^{0.75}$ |
| Efficiency of conversion MP to NP for maintenance (EMP <sub>M</sub> ) | $0.705 \pm 0.02$                          |
| MP requirement for maintenance (MP <sub>M</sub> ), g/d                | $\text{NP}_M / \text{EMP}_M$              |
| Net protein requirement for lactation (NP <sub>L</sub> ), g/d         | $\text{MY} \times \text{MTP}$             |
| Efficiency of the use MP to NP for lactation (EMP <sub>L</sub> )      | $0.705 \pm 0.02$                          |
| MP requirement for lactation (MP <sub>L</sub> ), g/d                  | $\text{NP}_L / \text{EMP}_L$              |
| Total MP requirement, g/d                                             | $\text{MP}_M + \text{MP}_L$               |

<sup>1</sup>MY = milk yield (kg/d); MTP = milk true protein (g/kg milk) = milk CP  $\times$  0.955 (g/kg milk).

(2018), and NASEM (2021) models underestimated the MP requirement and had a higher mean bias ( $367 \pm 30.1$  to  $859 \pm 33.4$  g/d;  $P < 0.01$ ; 43.0–77.5% MSPE), lower CCC (0.22–0.52; Table 5), and higher RMSPE

(24.3–42.7% observed MP; Figure 5) than our proposed model (Figures 3 and 4).

Compared with our proposed model, all external evaluated models had similar precision ( $\rho = 0.63$ –0.72;



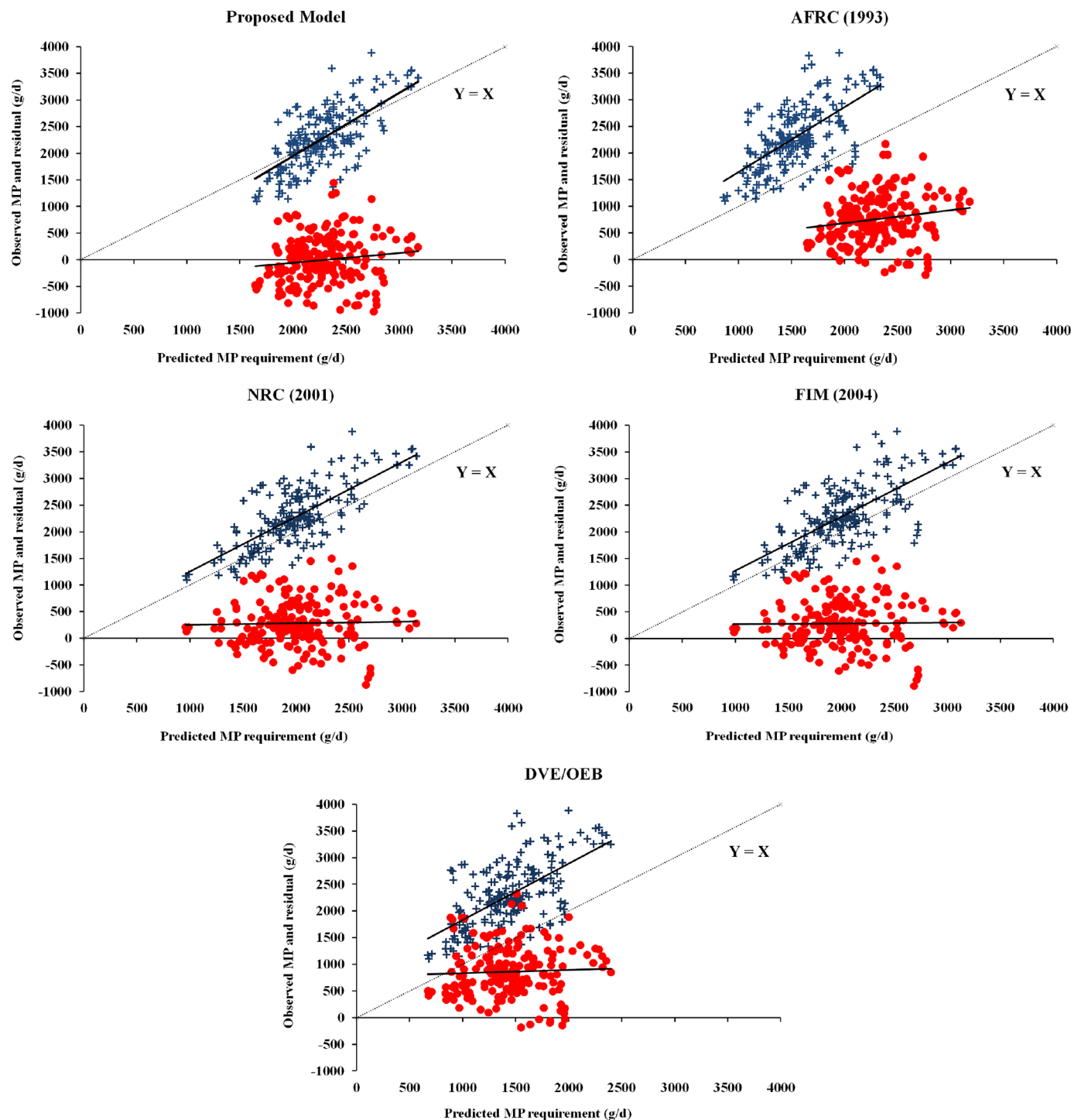
**Figure 2.** Relationship between the sum of N total fecal and urinary excretion (NFU; g of N/kg of  $\text{BW}^{0.75}$ ) and N intake (g of N/kg of  $\text{BW}^{0.75}$ ) by each N balance trial. The dotted lines are the observed values in each balance trial ( $n = 199$  N balance trials). The model is presented in Figure 1. Variance analysis of the MP residual was obtained from a mixed model with random effect of study and variance component structure.



**Table 5.** Summary of statistical measures to assess the adequacy of the proposed and external<sup>1</sup> predictive models of MP requirement for lactating dairy cows using regression between observed (Y) and predicted MP requirement

| Item <sup>2</sup>                           | Proposed model <sup>3</sup> | AFRC (1993)     | NRC (2001)      | FIM (2004) <sup>4</sup> | DVE/<br>OEB <sub>2010</sub> <sup>5</sup> | CSIRO (2007)    | CNCPS 6.5 <sup>6</sup> | INRA (2018)     | NASEM (2021)    |
|---------------------------------------------|-----------------------------|-----------------|-----------------|-------------------------|------------------------------------------|-----------------|------------------------|-----------------|-----------------|
| Observed MP (Y), g/d                        | 2,294                       | 2,294           | 2,294           | 2,294                   | 2,294                                    | 2,294           | 2,294                  | 2,294           | 2,294           |
| Predicted MP requirement (X), g/d           | 2,298                       | 1,535           | 2,012           | 2,011                   | 1,435                                    | 1,766           | 2,002                  | 1,927           | 1,797           |
| N                                           | 192                         | 192             | 192             | 192                     | 192                                      | 192             | 192                    | 192             | 192             |
| Residuals analysis <sup>7</sup> (Y – X)     |                             |                 |                 |                         |                                          |                 |                        |                 |                 |
| Intercept ( $\beta_0$ ; mean bias)          | $-4.0 \pm 32.8$             | $759 \pm 33.1$  | $282 \pm 30.7$  | $283 \pm 31.1$          | $859 \pm 33.4$                           | $528 \pm 31.6$  | $292 \pm 30.3$         | $367 \pm 30.1$  | $497 \pm 32.1$  |
| P-value ( $H_0$ , $\beta_0 = \text{zero}$ ) | 0.90                        | <0.01           | <0.01           | <0.01                   | <0.01                                    | <0.01           | <0.01                  | <0.01           | <0.01           |
| Slope ( $\beta_1$ ; slope bias)             | $0.19 \pm 0.10$             | $0.22 \pm 0.11$ | $0.03 \pm 0.08$ | $0.01 \pm 0.08$         | $0.06 \pm 0.09$                          | $0.11 \pm 0.09$ | $0.04 \pm 0.07$        | $0.10 \pm 0.08$ | $0.12 \pm 0.09$ |
| P-value ( $H_0$ , $\beta_1 = \text{zero}$ ) | 0.07                        | 0.04            | 0.72            | 0.88                    | 0.54                                     | 0.20            | 0.60                   | 0.19            | 0.18            |
| Root MSPE, g/d                              | 459                         | 893             | 512             | 516                     | 980                                      | 689             | 512                    | 558             | 670             |
| Partition of MSPE, %                        |                             |                 |                 |                         |                                          |                 |                        |                 |                 |
| Error due to mean bias                      | 0.0                         | 72.3            | 30.6            | 30.2                    | 77.5                                     | 43.6            | 32.7                   | 43.0            | 55.1            |
| Error due to slope not equal to 1           | 5.8                         | 1.4             | 0.3             | 0.3                     | 0.1                                      | 0.7             | 0.5                    | 1.7             | 1.1             |
| Random error                                | 94.1                        | 26.3            | 69.1            | 69.5                    | 22.4                                     | 55.7            | 66.8                   | 55.3            | 43.8            |
| CCC (0–1)                                   | 0.55                        | 0.23            | 0.57            | 0.55                    | 0.22                                     | 0.38            | 0.57                   | 0.52            | 0.39            |
| $\rho$ (0–1)                                | 0.65                        | 0.64            | 0.70            | 0.69                    | 0.63                                     | 0.68            | 0.71                   | 0.72            | 0.67            |
| $C_b$ (0–1)                                 | 0.84                        | 0.36            | 0.81            | 0.80                    | 0.35                                     | 0.56            | 0.80                   | 0.72            | 0.58            |

<sup>1</sup>External models are described in Supplemental Material S3 (<https://data.mendeley.com/datasets/z6pdk5pyg9>; Silva and Oliveira, 2022b).<sup>2</sup>MSPE = mean squared prediction error; CCC = concordance correlation coefficient;  $\rho$  = correlation coefficient estimate (precision);  $C_b$  = bias correction factor (accuracy).<sup>3</sup>Table 4.<sup>4</sup>FIM = Feed into Milk (Thomas, 2004).<sup>5</sup>DVE/OEB<sub>2010</sub> = the Dutch protein evaluation system for ruminants (Van Duinkerken et al., 2011).<sup>6</sup>Fox et al. (2004) and Van Amburgh et al. (2015).<sup>7</sup>Regression between residual (Y – X) and predicted MP centralized to its mean value (St-Pierre, 2003), adjusted for random effect of study. (Values  $\pm$  SE.)



**Figure 3.** Relationship between observed (blue +) and residual (observed – predicted; red circles) MP, with predicted MP values for lactating dairy cows using the proposed model (Table 4), AFRC (1993), NRC (2001), Feed into Milk (FIM; Thomas, 2004), and the Dutch protein evaluation system for ruminants (DVE/OEB<sub>2010</sub>; Van Duinkerken et al., 2011) models (descriptions of the external models are available in Supplemental Material S3, <https://data.mendeley.com/datasets/z6pdk5pyg9>; Silva and Oliveira, 2022b).

Table 5), but only the NRC (2001), Feed into Milk (Thomas, 2004), and CNCPS 6.5 (Fox et al., 2004; Van Amburgh et al., 2015) models predicted the MP

requirement with similar visual graphic adjustment between observed and predicted values (Figures 3 and 4) and had similar accuracy (CCC = 0.55–0.57; RM-

**Table 6.** ANOVA component to identify potential animal performance and dietary variables, and digesta sampling method affecting the MP residual (observed minus predicted) of the proposed model of MP requirement for maintenance and lactation

| Effect                                       | Multivariable model (n = 192 observations) <sup>1</sup> |         |                                 |
|----------------------------------------------|---------------------------------------------------------|---------|---------------------------------|
|                                              | Estimate (SE)                                           | P-value | Variance (% total) <sup>2</sup> |
| DMI, kg/d                                    | 61.1 (12.1)                                             | <0.01   | 15.6                            |
| Milk protein yield, kg/d                     | -558 (195)                                              | <0.01   | 3.1                             |
| BW, kg                                       | -0.59 (0.47)                                            | 0.21    | 0.2                             |
| NDF diet, g/kg DM                            | -1.48 (0.49)                                            | <0.01   | 3.3                             |
| CP diet, g/kg DM                             | 8.22 (1.81)                                             | <0.01   | 10.1                            |
| OM total-tract digestibility                 | -3.80 (0.75)                                            | <0.01   | 10.7                            |
| Duodenal versus omasal sampling <sup>3</sup> | 43.6 (75.6)                                             | 0.57    | 0.002                           |
| Random effect of study                       | —                                                       | —       | 2.6                             |
| Random residual                              | —                                                       | —       | 54.4                            |

<sup>1</sup>Y = MP residual (observed minus predicted) from the proposed model (Table 4).

<sup>2</sup>Variance analysis of the MP residual was obtained from a mixed model with random effect of study and variance component structure.

<sup>3</sup>Duodenal sampling = 0; omasal sampling = 1.

SPE = 22.3–22.5% observed MP; Table 5; Figure 5). However, these 3 external models predicted MP with higher error owing to mean bias (30.2–32.7% MSPE) compared with the proposed model (Table 5; Figures 3 and 4).

## DISCUSSION

We developed a new factorial protein requirement system for dairy cow maintenance and lactation from a meta-analysis of a large and comprehensive database of N balance trials. A model for  $NP_M$  and fixed  $EMP_L$  ( $EMP_L = 0.705 \pm 0.020$ ) was proposed and evaluated. In addition, we compared the adequacy of our system and several external models for predicting MP requirements of lactating dairy cows.

Overall, our findings indicate that our proposed fixed- $EMP_L$  model predicts the MP requirement with better accuracy than the AFRC (1993), CSIRO (2007), DVE/OEB<sub>2010</sub> (Van Duinkerken et al., 2011), INRA (2018), and NASEM (2021) models. Only the NRC (2001), Feed into Milk (Thomas, 2004), and CNCPS 6.5 (Fox et al., 2004; Van Amburgh et al., 2015) models predicted MP requirements with similar accuracy to our model, but their predictions had a higher error than our proposed fixed- $EMP_L$  model due to mean bias.

Our study also showed that the new NASEM (2021) model did not improve the MP requirement compared with the previous edition (NRC, 2001) and other models, such as CNCPS 6.5 and INRA (2018). We should highlight that the NASEM (2021) committee improved the prediction of milk protein yield from absorbed EAA supply, digestible energy intake, digestible NDF in diet, and BW compared with NRC (2001); however, an evaluation of its proposed model to predict MP re-

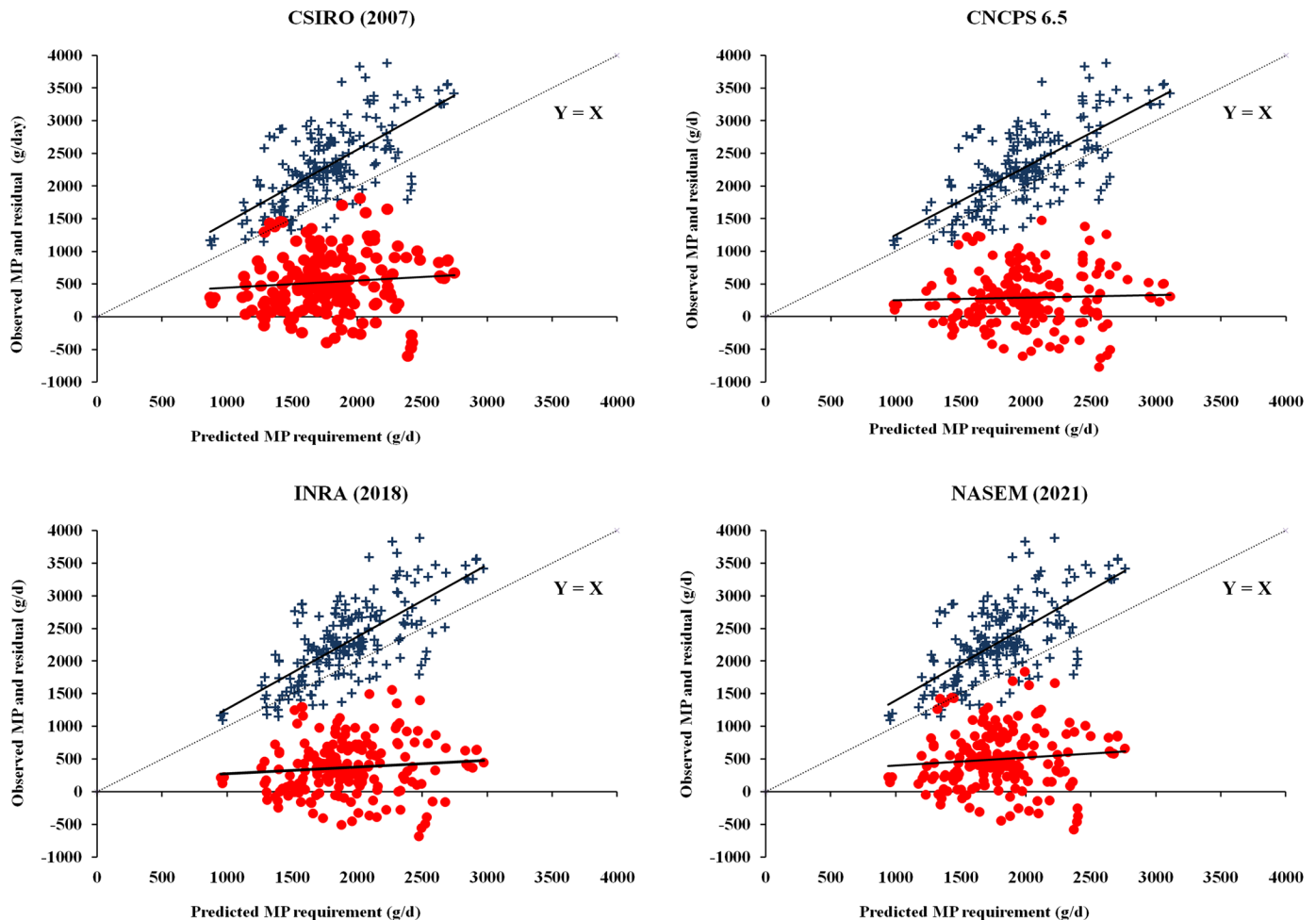
quirement against observed MP seems not to have been reported in NASEM (2021).

All external models predicted the MP requirement with higher error than our proposed model, owing to mean bias. However, prediction mean bias may be partially solved through incorporation of an empirical constant of mean bias adjustment. For example, the underestimate of the MP requirement of the NASEM (2021) model (mean bias =  $497 \pm 32.1$  g of MP/d and 55.1% of MSPE) and INRA (2018) (mean bias =  $367 \pm 30.1$  g of MP/d and 43.0% of MSPE) can be partially corrected by a further empirical incorporation of a positive constant of mean adjustment (i.e., intercept) in these models.

Dry matter intake, OM total-tract digestibility diet, and CP diet were the evaluated variables that most affected the linear prediction bias of our proposed model. In addition, we observed that the increase of DMI and CP diet increased MP residual, while the increase of OM total-tract digestibility reduced MP residual. These findings indicate the use of energy and protein supplies as inputs may further improve the MP prediction of our model.

## Protein Requirement for Maintenance

We assumed that  $NP_M$  represents the endogenous  $NFU \times 6.25$  (N to protein conversion factor). Scurf protein losses (skin and hair scaling) were not accounted for in  $NP_M$  in our system for two reasons: (1) absence of data in our data set and (2) scurf protein represents only 1.8% of the EPFU, calculated from Lapierre et al. (2020) and assuming dairy cows with 616 kg of BW, 21 kg/d DMI, and 34% NDF in DM diet (Table 1). To calculate the MP requirement for maintenance from



**Figure 4.** Relationship between observed (+) and residual (observed – predicted; circles) MP, with predicted MP values for lactating dairy cows using CSIRO (2007), CNCPS 6.5 (Fox et al., 2004; Van Amburgh et al., 2015), INRA (2018), and NASEM (2021) models (descriptions of the external model are available in Supplemental Material S3, <https://data.mendeley.com/datasets/z6pdk5pyg9>; Silva and Oliveira, 2022b).

$NP_M$ , we assumed that maintenance has the same MP efficiency for lactation; a similar approach was adopted for most of the external models evaluated in this study (Supplemental Material S3, <https://data.mendeley.com/datasets/z6pdk5pyg9>; Silva and Oliveira, 2022b).

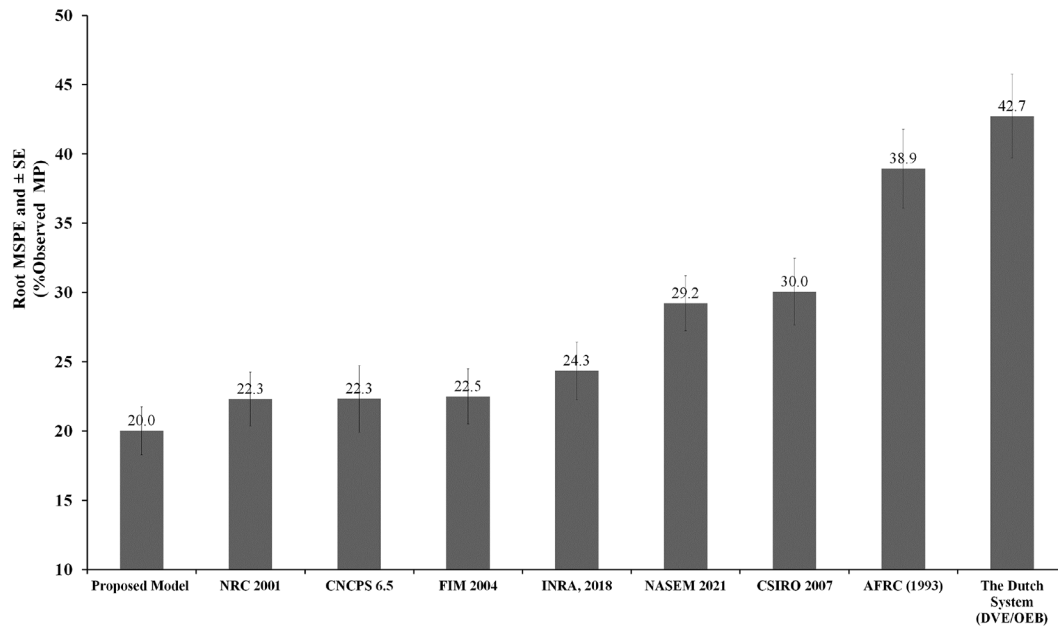
The endogenous NFU excretion was estimated from the intercept of a nonlinear meta-regression between N intake (g/kg of  $BW^{0.75}$ ; X) and total NFU (g/kg of  $BW^{0.75}$ ; Y). The model showed a strong correlation between N intake and NFU, and it supported our hypothesis that meta-analysis of N balance trials may provide a robust estimate of EPFU and  $NP_M$  from regression between N intake and NFU.

To the best of our knowledge, this study was the first to derive endogenous NFU to predict  $NP_M$  from a meta-analysis of N balance trials with lactating dairy cows. Several dairy cow nutrition models [NRC, 2001; Feed into Milk (Thomas, 2004); DVE/OEB<sub>2010</sub> (Van

Duinkerken, 2011); CNCPS 6.5 (Fox et al., 2004; Van Amburgh et al., 2015)] have adopted endogenous protein urinary ( $2.75 \times BW^{0.50}$ ; g/d), fecal [ $0.068 \times DMI \times \text{indigestible DM diet (\%)} \times 10$ ; g/d], or both types of excretion from Swanson (1977) meta-regression trials with nonlactating cattle. In addition, Swanson (1977) calculated EPFU from 16 low-protein and 70 low-N natural semisynthetic diets, respectively. Low or protein-free diets are a limited approach because feed intake is usually reduced when N is deficient, and the animals might adapt to the low-protein diets ingested; thus, the estimates obtained might not represent the fecal and urinary excretion under normal feeding conditions (Marini et al., 2008).

The NASEM (2021) committee adopted the Lapierre et al. (2020) assumptions, and its model is based on an admittedly scarce literature review of dairy cows to predict true protein, endogenous urinary (g/d;  $0.331 \times$





**Figure 5.** Values of root mean square prediction error (MSPE) and the standard error of the MP requirement (% observed MP) predictive for lactating dairy cows from our proposed model (Table 4), and AFRC (1993), NRC (2001), Feed into Milk (FIM; Thomas, 2004), the Dutch protein evaluation system for ruminants (DVE/OEB<sub>2010</sub>; Van Duinkerken et al., 2011), CSIRO (2007), CNCPS 6.5 (Fox et al., 2004; Van Amburgh et al., 2015), INRA (2018), and NASEM (2021) models (descriptions of the external models are available in Supplemental Material S3, <https://data.mendeley.com/datasets/z6pdk5pyg9>; Silva and Oliveira, 2022b).

BW<sub>kg</sub>) and fecal [g/d;  $(8.50 + 0.10 \times \text{NDF diet}_{\%DM}) \times \text{DMI}_{kg}$ ]. Similarly, the INRA (2018) committee adopted the Lapierre et al. (2016) recommendation to predict net protein endogenous urinary (g/d;  $0.312 \times \text{BW}_{kg}$ ), but the INRA (2018) committee developed their model based on meta-regression between undigested OM and N endogenous fecal from 214 experiments.

Our model estimated NP<sub>M</sub> of 668 to 860 g/d for cows of 500 to 700 kg of BW, while Swanson (1977) estimated 539 to 880 g/d (assuming DMI of 15.9 and 26.9 kg/d), CNCPS 6.5 (Fox et al., 2004; Van Amburgh et al., 2015) 542 to 883 g/d, INRA (2018) 404 to 634 g/d (assuming undigestible OM in diet of 330 g/kg DM), and NASEM (2021) 379 to 537 g/d (assuming 25 and 45% NDF in DM diet; Supplemental Material S3, <https://data.mendeley.com/datasets/z6pdk5pyg9>; Silva and Oliveira, 2022b). Our proposed model was developed from the analysis of N balance of lactating dairy cows, while these external nutrition models adopted equations from a limited lactating dairy cow data set. Therefore, the higher N intake and excretion per body mass of lactating dairy cows compared with nonlactating cattle might partially explain the higher estimated NP<sub>M</sub> values in our model compared with these external models. In addition, we estimated NP<sub>M</sub> from production-level diets, while Swanson (1977) used low or protein-free diet data.

The NRC (2001) and CNCPS 6.5 (Fox et al., 2004; Van Amburgh et al., 2015) also adopted the Swanson (1977) equation to estimate endogenous urinary protein excretion and scurf protein. However, the adoption of an adapted approach from Swanson (1977) to estimate endogenous fecal excretion seems to have contributed to improving the MP requirement prediction of the NRC (2001) and CNCPS 6.5 models. The Feed into Milk model (Thomas, 2004) adopted the NRC (2001) model to predict MP requirement for maintenance.

### MP Efficiency for Lactation

Most dairy cattle nutrition committees have also adopted fixed-EMP<sub>L</sub> values to predict MP<sub>L</sub> from NP<sub>L</sub>, with EMP<sub>L</sub> ranging from 0.67 to 0.70 (AFRC, 1993; NRC, 2001; CSIRO, 2007; Thomas, 2004; Van Amburgh et al., 2015; INRA, 2018; NASEM 2021). INRA (2018) and NASEM (2021) have adopted variable-EMP<sub>L</sub> models to predict MP supply and milk protein yield, but only fixed-EMP<sub>L</sub> values of 0.67 and 0.69 were used as targets to predict the MP<sub>L</sub> requirement from NP<sub>L</sub>.

Only the DVE/OEB<sub>2010</sub> (Van Duinkerken et al., 2011) and NorFor (2011) models have used variable-EMP<sub>L</sub> to predict MP<sub>L</sub> from NP<sub>L</sub> requirements. The

DVE/OEB<sub>2010</sub> model has used the milk protein to energy ratio as input for predicting EMP<sub>L</sub>, and NorFor (2011) used the ratio of milk production to milk energy. In DVE/OEB<sub>2010</sub>, EMP<sub>L</sub> decreases with milk protein yield (Van Duinkerken et al., 2011), whereas in NorFor (2011), neither the milk protein yield nor the feeding level affects EMP<sub>L</sub>. Although variable-EMP<sub>L</sub>-based models are biochemically more realistic, the DVE/OEB<sub>2010</sub> model predicted MP requirement with less accuracy than fixed-EMP<sub>L</sub>-based models. These results indicate that developing accurate variable-EMP<sub>L</sub>-based models remains a challenge.

The lower EMP<sub>L</sub> of *B. taurus* × *B. indicus* crossbred dairy cows may be associated with lower ME efficiency for lactation compared with *B. taurus* × *B. taurus* (Oliveira, 2015). Differences in homeostasis and homeorhetic regulations for lactation and body reserve have been proposed as causal hypotheses for this difference in ME efficiency for lactation (Oliveira, 2015). As milk energy efficiency (energy in milk/digestible energy intake) and N milk efficiency (N milk/digestible N intake) are also positively correlated (Phuong et al., 2013), it is possible to hypothesize that the lower EMP<sub>L</sub> of *B. taurus* × *B. indicus* is associated with lower ME efficiency.

We calculated EMP<sub>L</sub> as a proportion of N intake that was secreted in milk, discounting NFU (Equation 2). NRC (2001), CNCPS 6.5, and INRA (2018) have adopted an EMP<sub>L</sub> of 0.67, which was originally proposed by Vérité et al. (1979) from the equation: EMP<sub>L</sub> = observed milk protein yield/(estimated MP intake – estimated MP requirement for maintenance). NASEM (2021) has adopted a variable-EMP<sub>L</sub>-based model to predict MP supply and milk protein yield. However, only a fixed and combined efficiency for use of MP maintenance and lactation (Eff<sub>MP</sub>) of 0.69 was proposed as a target to predict the MP<sub>L</sub> requirement from NP<sub>L</sub>, estimated from the equation of Lapierre et al. (2020): Eff<sub>MP</sub> = (estimated true protein scurf secretion + estimated true protein metabolic fecal secretion + observed milk protein yield from 921 treatment means of 216 studies)/(estimated MP supply – estimated true protein endogenous urinary excretion). Although we proposed a different approach to measure the EMP<sub>L</sub> in this study, our obtained EMP<sub>L</sub> value was similar to the values from most of the external evaluated models (Supplemental Material S3, <https://data.mendeley.com/datasets/z6pdk5pyg9>; Silva and Oliveira, 2022b). Therefore, differences in the quality of the MP requirement (maintenance + lactation) prediction between the evaluated models are more associated with differences in the maintenance requirement than lactation.

## Limitations

Our study has some limitations. First, specific phenomena that may potentially affect the estimation of maintenance requirements, such as the efficiency and use of endogenous protein and recycled urea N by rumen microbes (Lapierre et al., 2016), were not considered. Second, although our nonlinear equation showed strong goodness-of-fit to estimate EPFU from N intake, more factors may be involved in endogenous N fecal losses, such as NDF content and rate of fermentation of dietary carbohydrates (Marini et al., 2008). Third, our data set has insufficient *B. taurus* × *B. indicus* data (n = 26 treatment means) to compare with *B. taurus* (n = 705; mainly Holstein). Therefore, our finding that *B. taurus* × *B. indicus* dairy cows have lower EMP<sub>L</sub> than *B. taurus* (0.594 ± 0.045 vs. 0.705 ± 0.020) has limited robustness and needs to be confirmed in further investigations.

Finally, to evaluate and compare the adequacy of our predictive MP requirement system with external models, we calculated the observed MP from an independent trial data set for which microbial N and NANMN (as a proxy for RUP) ruminal outflows were measured (Equation 3). However, we had to assume fixed values of intestinal digestibilities for microbial protein (NASEM, 2021) and RUP (White et al., 2017). Microbial protein intestinal digestibility has been found to range from 57 to 87% according to bacterial/protozoa/fungal proportion and adopted method (Jouany, 1996; Larsen et al., 2001; Fonseca et al., 2014; Fessenden et al., 2017), while RUP-total AA intestinal digestibility has ranged from 52 to 94% according to feed, feed processing, and method (i.e., incubation of residue in digestive enzymes in vitro or in mobile bags inserted into the duodenum) (White et al., 2017). Therefore, more refined predictive models of intestinal digestibility for RUP and microbial protein are needed to reduce these evaluation biases in our protein requirement system and external models for lactating dairy cows.

## CONCLUSIONS

A new factorial system for accounting NP<sub>M</sub> and MP efficiency for lactation was developed from a meta-analysis of N balance trials of lactating dairy cows. We estimated the EPFU (NP<sub>M</sub>) from the intercept of a nonlinear equation between N intake and combined N fecal and urinary excretions. The proposed model provided a robust estimation of EPFU. We proposed a fixed-EMP<sub>L</sub> calculated as a proportion of the N metabolizable (N intake minus N fecal and urinary) that was secreted in milk. Our system predicted the MP

requirement for lactation and maintenance with higher accuracy than several North American and European dairy cattle nutrition models, including INRA (2018) and NASEM (2021). Only the NRC (2001), Feed into Milk, and CNCPS 6.5 models presented similar accuracy for predicting the MP requirement, but they had higher error than our fixed-EMP<sub>L</sub>-based model owing to mean bias. Our system may contribute to improve the MP requirement prediction for maintenance and lactation. However, more refined predictive models of intestinal digestibility for RUP and microbial protein are still needed to reduce the evaluation biases in our system and external models for predicting NP and MP requirements of dairy cows.

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